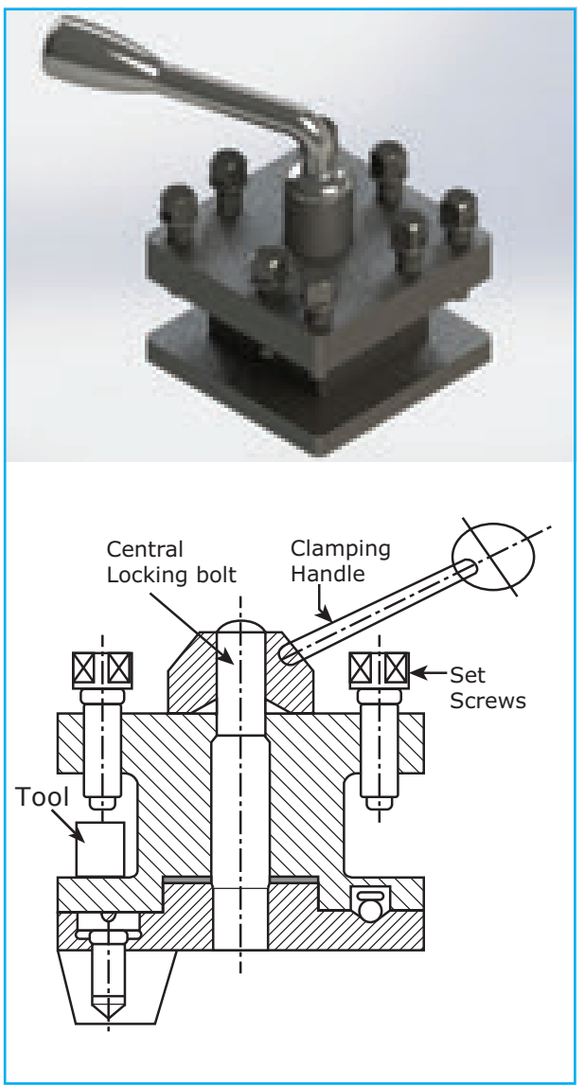
The height of the tool point can be adjusted by moving the rocker front and back. As the tool is clamped by only one clamping screw, it is not preferable for heavy duty works.

Four way tool post

Four way tool post is known as Turret tool post. Four different types of tool can be placed and clamped at a time. The required tool can be turned towards the workpiece for turning process. As the required tools fitted ready in position, the turning work is easier and faster. This type of tool post is used in heavy duty lathes.



Four way tool post

1.5 Feed Mechanism

By using various feed mechanism, the rotation of leadscrew and feed rod direction can be changed and so the direction of carriage and cross slide movements can also be changed.

There are so many feed mechanisms used in lathe, among them the important mechanisms are discussed below.

1.6 Tumbler Gear

A set of two small gears are placed just below the spindle gear and connected to the spindle gear in the lathe. The set of gears is known as tumbler gear.

The set of two small tumbler gears are fitted in a bracket. The bracket position can be adjusted into three positions such as neutral, upwards and downwards. By adjusting the position of tumbler gear, the rotation of leadscrew and feed rods direction can be changed.

Neutral Position

In this position, the tumbler gears lever is in the middle position hence, the tumbler gears are not engaged with spindle gear and the intermediate gears and so, automatic feed is not received.

Forward position

In this position, the tumbler gears lever is shifted upward position. Here only one tumbler gear is in contact with the spindle gear, hence, the leadscrew rotates in the same direction of the spindle rotation. The carriage moves towards headstock.

Reverse Position

In this position, the tumbler gears lever is pushed downward position. Here

XII BME_English version CHAPTER 01.indd 7 1/8/2020 1:45:52 PM